DS2 · Topic 3.2

Sagot sa DS2 ni Seb

Matrices: Complete Reviewer with Full Solutions

Every example, exercise, and problem set from the *3.2 Matrices* slides, solved step by step and explained in simple words.

WHAT'S INSIDE

1. **What is a Matrix?** · definition, size, entries, types
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How to review with this: ① Read the 💡 Idea box. ② Follow the worked solution line by line. ③ Cover the answer box and solve it yourself. ④ Finish with the cheat sheet on the last pages.

1 What is a Matrix?

💡 **THE IDEA (IN PLAIN WORDS)**

A **matrix** is just a **rectangle of numbers** arranged in **rows** (horizontal ↔) and **columns** (vertical ↕). Think of it like a seating chart or a spreadsheet table.

The **size** (also called *order* or *dimension*) of a matrix is written as **rows × columns**, read as "*m by n*". **Rows always come first.**

✅ **TIP**

Memory trick: **RC**, like *Roman Catholic*. **R**ows first, then **C**olumns. This order is used for size AND for entry positions.

Example from the slides: a "2 by 3" matrix

$$\begin{bmatrix} 3 & 5 & 12 \\ -1 & -7 & 4 \end{bmatrix} \rightarrow \textbf{2 rows, 3 columns} \rightarrow \text{size } \mathbf{2\times3}$$

Naming each entry: a_{ij}

Each number inside is called an **entry** (or element). We name it $\mathbf{a_{ij}}$, where **i = row number** and **j = column number**.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \text{ an } \mathbf{m \times n} \text{ matrix}$$

Using the 2×3 example above:

- a_{12} = row 1, column 2 = **5**
- a_{21} = row 2, column 1 = **−1**
- a_{23} = row 2, column 3 = **4**

Types of matrices (by size)

Name	Size	Example	Meaning
Row matrix	1 × n	$\begin{bmatrix} 3 & 7 & 2 \end{bmatrix}$	Only **one row**. Often used as a vector.
Column matrix	m × 1	$\begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$	Only **one column**. Often used as a vector (like the x, y, z column later).

Square matrix	n × n	$\begin{bmatrix} 9 & 13 & 5 \\ 1 & 11 & 7 \\ 2 & 6 & 3 \end{bmatrix}$	**Same number of rows and columns.** Only square matrices have a determinant, an inverse, and powers.
Identity matrix (I)	n × n	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1s on the main diagonal, 0s everywhere else. It works like the number 1: A·I = A. Needed for the **inverse** topic.

📝 **NOTE ABOUT THE SLIDES**

Slide 7 lists the size of a square matrix as $m * n$. It should be **n × n** (or m × m), because a square matrix has the **same** number of rows and columns.

2 Scalar Multiplication

💡 **THE IDEA (IN PLAIN WORDS)**

A **scalar** is just a regular number (like 4 or −3). To multiply a matrix by a scalar, **multiply EVERY entry** by that number. The size stays the same.

$$(cA)_{i,j} = c \cdot A_{i,j}$$

Slide example: find 4A

$$A = \begin{bmatrix} 3 & 5 & 12 \\ -1 & -7 & 4 \end{bmatrix}$$

Step 1. Multiply each entry by 4:

$$4A = \begin{bmatrix} 4(3) & 4(5) & 4(12) \\ 4(-1) & 4(-7) & 4(4) \end{bmatrix}$$

Step 2. Simplify each one:

FINAL ANSWER

$$4A = \begin{bmatrix} 12 & 20 & 48 \\ -4 & -28 & 16 \end{bmatrix}$$

✅ **TIP**

If the scalar is **negative**, every sign flips. Example: (−1)·A turns 3 into −3 and −7 into 7.

3 Addition & Subtraction of Matrices

💡 **THE IDEA (IN PLAIN WORDS)**

You can only add or subtract matrices that have the **SAME size**. Then just add (or subtract) the numbers that are in the **same position**: top-left with top-left, and so on.

$$(A \pm B)_{i,j} = A_{i,j} \pm B_{i,j}$$

Slide example: addition

$$\begin{bmatrix} 1 & -4 & 8 \\ 11 & \mathbf{2} & 24 \\ 12 & 4 & 1 \end{bmatrix} + \begin{bmatrix} -9 & 8 & 6 \\ 0 & \mathbf{15} & 2 \\ 3 & 14 & 0 \end{bmatrix}$$

Both are 3×3, so addition is allowed ✓. Add position by position (the highlighted pair shows one example: 2 + 15 = 17):

$$= \begin{bmatrix} 1+(-9) & -4+(8) & 8+(6) \\ 11+(0) & \mathbf{2+(15)} & 24+(2) \\ 12+(3) & 4+(14) & 1+(0) \end{bmatrix}$$

FINAL ANSWER

$$= \begin{bmatrix} -8 & 4 & 14 \\ 11 & \mathbf{17} & 26 \\ 15 & 18 & 1 \end{bmatrix}$$

Bonus practice: subtraction of the same matrices

Subtracting works the same way. Be careful with negatives: **minus a negative becomes plus**, e.g. 1 − (−9) = 1 + 9 = 10.

$$\begin{bmatrix} 1 & -4 & 8 \\ 11 & 2 & 24 \\ 12 & 4 & 1 \end{bmatrix} - \begin{bmatrix} -9 & 8 & 6 \\ 0 & 15 & 2 \\ 3 & 14 & 0 \end{bmatrix} = \begin{bmatrix} 1-(-9) & -4-(8) & 8-(6) \\ 11-(0) & 2-(15) & 24-(2) \\ 12-(3) & 4-(14) & 1-(0) \end{bmatrix}$$

FINAL ANSWER

$$= \begin{bmatrix} 10 & -12 & 2 \\ 11 & -13 & 22 \\ 9 & -10 & 1 \end{bmatrix}$$

⚠️ **COMMON MISTAKE**

Adding a 2×3 matrix to a 3×2 matrix is **NOT defined**. The sizes must match exactly.

4 Transpose of a Matrix

💡 THE IDEA (IN PLAIN WORDS)

The **transpose** A^T flips the matrix: **row 1 becomes column 1, row 2 becomes column 2**, and so on. An m×n matrix becomes an **n×m** matrix.

$$(A^T)_{i,j} = A_{j,i}$$

Other notations for transpose: A^{tr} or ${}^tA$.

Slide example

Watch the colors: each **row** on the left becomes a **column** on the right.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad (3\times2)$$

FINAL ANSWER

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \quad (2\times3)$$

- Row 1 of A: 1, 2 → becomes column 1 of A^T
- Row 2 of A: 3, 4 → becomes column 2
- Row 3 of A: 5, 6 → becomes column 3

✅ TIP

Transposing twice brings you back: $(A^T)^T = A$. You will use the transpose again in **Problem Set 5 #10** and in the **Adjugate** step when finding an inverse.

5 Multiplication of Matrices

💡 THE IDEA (IN PLAIN WORDS)

Matrix multiplication is **NOT** position-by-position. Each answer entry is made by taking a **ROW of the left matrix** and a **COLUMN of the right matrix**, multiplying their matching numbers, and **adding** the results. This is called the **dot product**.

Rule 1: When is it allowed? (Check the sizes first!)

$$A_{m\times\mathbf{n}} \cdot B_{\mathbf{n}\times p} = C_{m\times p}$$

- The **inner numbers** must be equal (columns of the left = rows of the right).
- The **outer numbers** give the size of the answer.

Rule 2: How to get each entry

$$[AB]_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}$$

In words: **entry in row i, column j** = (row i of A) · (column j of B).

> 📝 **NOTE ABOUT THE SLIDES**
> On slide 18, the last term of the formula is typed as $a_{i,1}b_{1,j}$. It should be $\mathbf{a_{i,n}b_{n,j}}$ (the last pair in the row and column).

> ⚠️ **COMMON MISTAKE**
> **Order matters!** In general A·B ≠ B·A. Sometimes A·B is defined but B·A is not defined at all.

Slide example

$$\begin{bmatrix} \mathbf{-9} & \mathbf{6} \\ 0 & 2 \\ 3 & 0 \end{bmatrix} \times \begin{bmatrix} \mathbf{1} & -4 & 5 \\ \mathbf{7} & 2 & 2 \end{bmatrix}$$

Left is 3×**2** Right is **2**×3 → Inner numbers match (2 = 2) ✓ Answer size = **3×3**

Example of one entry (highlighted): **Row 1** = (−9, 6) and **Column 1** = (1, 7): (−9)(1) + (6)(7) = −9 + 42 = **33**. Do the same for every row–column pair:

Entry	What to multiply	Work	=
c_{11}	Row 1 of L · Col 1 of R	(−9)(1) + (6)(7) = −9 + 42	**33**
c_{12}	Row 1 of L · Col 2 of R	(−9)(−4) + (6)(2) = 36 + 12	**48**
c_{13}	Row 1 of L · Col 3 of R	(−9)(5) + (6)(2) = −45 + 12	**−33**
c_{21}	Row 2 of L · Col 1 of R	(0)(1) + (2)(7) = 0 + 14	**14**
c_{22}	Row 2 of L · Col 2 of R	(0)(−4) + (2)(2) = 0 + 4	**4**
c_{23}	Row 2 of L · Col 3 of R	(0)(5) + (2)(2) = 0 + 4	**4**
c_{31}	Row 3 of L · Col 1 of R	(3)(1) + (0)(7) = 3 + 0	**3**
c_{32}	Row 3 of L · Col 2 of R	(3)(−4) + (0)(2) = −12 + 0	**−12**

c_{33}	Row 3 of L · Col 3 of R	(3)(5) + (0)(2) = 15 + 0	**15**

FINAL ANSWER

$$= \begin{bmatrix} \mathbf{33} & 48 & -33 \\ 14 & 4 & 4 \\ 3 & -12 & 15 \end{bmatrix} \quad (3\times3)$$

Exercise 1: Compute the matrix product

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix}$$

Step 1: Check the sizes.

Left is 4×**3** Right is **3**×2 → Inner numbers match (3 = 3) ✓ Answer size = **4×2**

Step 2: Row × Column for each of the 8 entries.

Entry	What to multiply	Work	=
c_{11}	Row 1 of L · Col 1 of R	(1)(2) + (0)(1) + (4)(3) = 2 + 0 + 12	**14**
c_{12}	Row 1 of L · Col 2 of R	(1)(4) + (0)(1) + (4)(0) = 4 + 0 + 0	**4**
c_{21}	Row 2 of L · Col 1 of R	(2)(2) + (1)(1) + (1)(3) = 4 + 1 + 3	**8**
c_{22}	Row 2 of L · Col 2 of R	(2)(4) + (1)(1) + (1)(0) = 8 + 1 + 0	**9**
c_{31}	Row 3 of L · Col 1 of R	(3)(2) + (1)(1) + (0)(3) = 6 + 1 + 0	**7**
c_{32}	Row 3 of L · Col 2 of R	(3)(4) + (1)(1) + (0)(0) = 12 + 1 + 0	**13**
c_{41}	Row 4 of L · Col 1 of R	(0)(2) + (2)(1) + (2)(3) = 0 + 2 + 6	**8**
c_{42}	Row 4 of L · Col 2 of R	(0)(4) + (2)(1) + (2)(0) = 0 + 2 + 0	**2**

FINAL ANSWER

$$= \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix} \quad (4\times2)$$

6 Power of a Matrix

💡 **THE IDEA (IN PLAIN WORDS)**

A^2 means **A · A** (matrix multiplication of A with itself). $A^3 = A \cdot A \cdot A = A^2 \cdot A$.

- Only **square** matrices can be raised to a power. Why? For A·A, the inner numbers must match: a 2×3 matrix gives (2×**3**)(**2**×3), which is not allowed.

⚠️ **COMMON MISTAKE**

Do **NOT** just square each number! For the example below, $\begin{bmatrix} 4 & 36 \\ 25 & 9 \end{bmatrix}$ is **WRONG**.

Slide example

$$\begin{bmatrix} 2 & 6 \\ 5 & 3 \end{bmatrix}^2 = \begin{bmatrix} 2 & 6 \\ 5 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 6 \\ 5 & 3 \end{bmatrix}$$

A is 2×**2** A is **2**×2 → Inner numbers match (2 = 2) ✓ Answer size = **2×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A · Col 1 of A	(2)(2) + (6)(5) = 4 + 30	**34**
c_{12}	Row 1 of A · Col 2 of A	(2)(6) + (6)(3) = 12 + 18	**30**
c_{21}	Row 2 of A · Col 1 of A	(5)(2) + (3)(5) = 10 + 15	**25**
c_{22}	Row 2 of A · Col 2 of A	(5)(6) + (3)(3) = 30 + 9	**39**

FINAL ANSWER

$$\begin{bmatrix} 2 & 6 \\ 5 & 3 \end{bmatrix}^2 = \begin{bmatrix} 34 & 30 \\ 25 & 39 \end{bmatrix}$$

7 Exercise 2: Are these multiplications defined? If so, compute.

$$A = \begin{bmatrix} 1 & 2 \\ -2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -4 & 5 \\ 2 & 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 1 \\ 4 & 0 \\ 1 & 6 \end{bmatrix}$$

First, write the size of each: A is **2×2**, B is **2×3**, C is **3×2**. Then check (left columns) = (right rows) for each item.

Item	Sizes	Result
1. A·B	(2×2)(2×3)	**Defined** → 2×3
2. B·A	(2×3)(2×2)	**Not defined**
3. C·A	(3×2)(2×2)	**Defined** → 3×2
4. A·C	(2×2)(3×2)	**Not defined**
5. B·C	(2×3)(3×2)	**Defined** → 2×2
6. A·A	(2×2)(2×2)	**Defined** → 2×2
7. B·B	(2×3)(2×3)	**Not defined**

✅ **TIP**

Quick trick: write the sizes side by side, like (2×**2**)(**2**×3). If the two middle numbers are the same, it's defined, and the two outer numbers give the answer's size.

1. A·B

A is 2×**2** B is **2**×3 → Inner numbers match (2 = 2) ✓ Answer size = **2×3**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A · Col 1 of B	(1)(3) + (2)(2) = 3 + 4	**7**
c_{12}	Row 1 of A · Col 2 of B	(1)(−4) + (2)(1) = −4 + 2	**−2**
c_{13}	Row 1 of A · Col 3 of B	(1)(5) + (2)(0) = 5 + 0	**5**
c_{21}	Row 2 of A · Col 1 of B	(−2)(3) + (4)(2) = −6 + 8	**2**
c_{22}	Row 2 of A · Col 2 of B	(−2)(−4) + (4)(1) = 8 + 4	**12**

c_{23}	Row 2 of A · Col 3 of B	$(-2)(5) + (4)(0) = -10 + 0$	**−10**

FINAL ANSWER

$$A \cdot B = \begin{bmatrix} 7 & -2 & 5 \\ 2 & 12 & -10 \end{bmatrix}$$

2. B·A

B is 2×**3** A is **2**×2 → Inner numbers do NOT match (3 ≠ 2) ✗ **NOT DEFINED**

ANSWER

B·A is NOT defined: columns of B (3) ≠ rows of A (2).

3. C·A

C is 3×**2** A is **2**×2 → Inner numbers match (2 = 2) ✓ Answer size = **3×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of C · Col 1 of A	$(3)(1) + (1)(-2) = 3 - 2$	**1**
c_{12}	Row 1 of C · Col 2 of A	$(3)(2) + (1)(4) = 6 + 4$	**10**
c_{21}	Row 2 of C · Col 1 of A	$(4)(1) + (0)(-2) = 4 + 0$	**4**
c_{22}	Row 2 of C · Col 2 of A	$(4)(2) + (0)(4) = 8 + 0$	**8**
c_{31}	Row 3 of C · Col 1 of A	$(1)(1) + (6)(-2) = 1 - 12$	**−11**
c_{32}	Row 3 of C · Col 2 of A	$(1)(2) + (6)(4) = 2 + 24$	**26**

FINAL ANSWER

$$C \cdot A = \begin{bmatrix} 1 & 10 \\ 4 & 8 \\ -11 & 26 \end{bmatrix}$$

4. A·C

A is 2×**2** C is **3**×2 → Inner numbers do NOT match (2 ≠ 3) ✗ **NOT DEFINED**

ANSWER

A·C is NOT defined: columns of A (2) ≠ rows of C (3).

5. B·C

B is 2×**3** C is **3**×2 → Inner numbers match (3 = 3) ✓ Answer size = **2×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of B · Col 1 of C	(3)(3) + (−4)(4) + (5)(1) = 9 − 16 + 5	**−2**
c_{12}	Row 1 of B · Col 2 of C	(3)(1) + (−4)(0) + (5)(6) = 3 + 0 + 30	**33**
c_{21}	Row 2 of B · Col 1 of C	(2)(3) + (1)(4) + (0)(1) = 6 + 4 + 0	**10**
c_{22}	Row 2 of B · Col 2 of C	(2)(1) + (1)(0) + (0)(6) = 2 + 0 + 0	**2**

FINAL ANSWER

$$B\cdot C = \begin{bmatrix} -2 & 33 \\ 10 & 2 \end{bmatrix}$$

6. A·A

A is 2×**2** A is **2**×2 → Inner numbers match (2 = 2) ✓ Answer size = **2×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A · Col 1 of A	(1)(1) + (2)(−2) = 1 − 4	**−3**
c_{12}	Row 1 of A · Col 2 of A	(1)(2) + (2)(4) = 2 + 8	**10**
c_{21}	Row 2 of A · Col 1 of A	(−2)(1) + (4)(−2) = −2 − 8	**−10**
c_{22}	Row 2 of A · Col 2 of A	(−2)(2) + (4)(4) = −4 + 16	**12**

FINAL ANSWER

$$A\cdot A = \begin{bmatrix} -3 & 10 \\ -10 & 12 \end{bmatrix}$$

7. B·B

B is 2×**3** B is **2**×3 → Inner numbers do NOT match (3 ≠ 2) ✗ **NOT DEFINED**

ANSWER

B·B is NOT defined: columns of B (3) ≠ rows of B (2).

8 Problem Set 5 (All 10 items solved)

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 3 & 2 \\ 6 & 2 & 1 \end{bmatrix} \quad C = \begin{bmatrix} -5 & 1 \\ 2 & -3 \\ 4 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 2 \\ 5 & 3 & -3 \\ 6 & -1 & 2 \end{bmatrix}$$

Sizes: A is **2×2**, B is **2×3**, C is **3×2**, D is **3×3**.

#	Problem	Answer at a glance
1	A·B	$\begin{bmatrix} 17 & 7 & 4 \\ 39 & 17 & 10 \end{bmatrix}$
2	C·A	$\begin{bmatrix} -2 & -6 \\ -7 & -8 \\ 4 & 8 \end{bmatrix}$
3	A^2	$\begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$
4	C^2	**Not defined** (C is not square)
5	B·C	$\begin{bmatrix} -11 & -4 \\ -22 & 0 \end{bmatrix}$
6	4A·(−3)B	$\begin{bmatrix} -204 & -84 & -48 \\ -468 & -204 & -120 \end{bmatrix}$
7	B·2C	$\begin{bmatrix} -22 & -8 \\ -44 & 0 \end{bmatrix}$
8	det(D)	**−43**
9	B·A	**Not defined**
10	A^T·B	$\begin{bmatrix} 23 & 9 & 5 \\ 34 & 14 & 8 \end{bmatrix}$

1. A · B

A is 2×**2** B is **2**×3 → Inner numbers match (2 = 2) ✓ Answer size = **2×3**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A · Col 1 of B	(1)(5) + (2)(6) = 5 + 12	**17**
c_{12}	Row 1 of A · Col 2 of B	(1)(3) + (2)(2) = 3 + 4	**7**
c_{13}	Row 1 of A · Col 3 of B	(1)(2) + (2)(1) = 2 + 2	**4**

c_{21}	Row 2 of A · Col 1 of B	(3)(5) + (4)(6) = 15 + 24	**39**
c_{22}	Row 2 of A · Col 2 of B	(3)(3) + (4)(2) = 9 + 8	**17**
c_{23}	Row 2 of A · Col 3 of B	(3)(2) + (4)(1) = 6 + 4	**10**

FINAL ANSWER

$$A \cdot B = \begin{bmatrix} 17 & 7 & 4 \\ 39 & 17 & 10 \end{bmatrix}$$

2. C · A

C is 3×**2** A is **2**×2 → Inner numbers match (2 = 2) ✓ Answer size = **3×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of C · Col 1 of A	(−5)(1) + (1)(3) = −5 + 3	**−2**
c_{12}	Row 1 of C · Col 2 of A	(−5)(2) + (1)(4) = −10 + 4	**−6**
c_{21}	Row 2 of C · Col 1 of A	(2)(1) + (−3)(3) = 2 − 9	**−7**
c_{22}	Row 2 of C · Col 2 of A	(2)(2) + (−3)(4) = 4 − 12	**−8**
c_{31}	Row 3 of C · Col 1 of A	(4)(1) + (0)(3) = 4 + 0	**4**
c_{32}	Row 3 of C · Col 2 of A	(4)(2) + (0)(4) = 8 + 0	**8**

FINAL ANSWER

$$C \cdot A = \begin{bmatrix} -2 & -6 \\ -7 & -8 \\ 4 & 8 \end{bmatrix}$$

3. $A^2 = A \cdot A$

A is 2×**2** A is **2**×2 → Inner numbers match (2 = 2) ✓ Answer size = **2×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A · Col 1 of A	(1)(1) + (2)(3) = 1 + 6	**7**
c_{12}	Row 1 of A · Col 2 of A	(1)(2) + (2)(4) = 2 + 8	**10**
c_{21}	Row 2 of A · Col 1 of A	(3)(1) + (4)(3) = 3 + 12	**15**

c_{22}	Row 2 of A · Col 2 of A	$(3)(2) + (4)(4) = 6 + 16$	**22**

FINAL ANSWER

$$A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

4. $C^2 = C \cdot C$

C is 3×**2** C is **3**×2 → Inner numbers do NOT match (2 ≠ 3) ✗ **NOT DEFINED**

ANSWER

C^2 is NOT defined. C is 3×2, which is not square, so (3×**2**)(**3**×2) doesn't match.

5. B · C

B is 2×**3** C is **3**×2 → Inner numbers match (3 = 3) ✓ Answer size = **2×2**

Entry	What to multiply	Work	=
c_{11}	Row 1 of B · Col 1 of C	$(5)(-5) + (3)(2) + (2)(4) = -25 + 6 + 8$	**−11**
c_{12}	Row 1 of B · Col 2 of C	$(5)(1) + (3)(-3) + (2)(0) = 5 - 9 + 0$	**−4**
c_{21}	Row 2 of B · Col 1 of C	$(6)(-5) + (2)(2) + (1)(4) = -30 + 4 + 4$	**−22**
c_{22}	Row 2 of B · Col 2 of C	$(6)(1) + (2)(-3) + (1)(0) = 6 - 6 + 0$	**0**

FINAL ANSWER

$$B \cdot C = \begin{bmatrix} -11 & -4 \\ -22 & 0 \end{bmatrix}$$

6. 4A · (−3)B

Shortcut: Scalars can be pulled out front and multiplied together: 4A · (−3)B = (4)(−3) · (A·B) = **−12 · (A·B)**. We already have A·B from #1.

$$-12 \cdot \begin{bmatrix} 17 & 7 & 4 \\ 39 & 17 & 10 \end{bmatrix} = \begin{bmatrix} -12(17) & -12(7) & -12(4) \\ -12(39) & -12(17) & -12(10) \end{bmatrix}$$

FINAL ANSWER

$$4A \cdot (-3)B = \begin{bmatrix} -204 & -84 & -48 \\ -468 & -204 & -120 \end{bmatrix}$$

Check (long way): first compute 4A and (−3)B, then multiply them:

$$4A = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix} \quad (-3)B = \begin{bmatrix} -15 & -9 & -6 \\ -18 & -6 & -3 \end{bmatrix}$$

e.g. row 1 · col 1 = (4)(−15) + (8)(−18) = −60 − 144 = −204 ✓. Same answer as the shortcut.

7. B · 2C

Shortcut: B · 2C = 2 · (B·C). We already have B·C from #5.

$$2 \cdot \begin{bmatrix} -11 & -4 \\ -22 & 0 \end{bmatrix} = \begin{bmatrix} 2(-11) & 2(-4) \\ 2(-22) & 2(0) \end{bmatrix}$$

FINAL ANSWER

$$B{\cdot}2C = \begin{bmatrix} -22 & -8 \\ -44 & 0 \end{bmatrix}$$

Check (long way):

$$2C = \begin{bmatrix} -10 & 2 \\ 4 & -6 \\ 8 & 0 \end{bmatrix}$$

Row 1 of B · col 1 of 2C = (5)(−10) + (3)(4) + (2)(8) = −50 + 12 + 16 = −22 ✓. Same answer.

8. det(D)

D is 3×3, so use the **diagonal (Sarrus) method** from the Determinant section. Label D as

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 5 & 3 & -3 \\ 6 & -1 & 2 \end{bmatrix}$$

Down diagonals (+)	Up diagonals (−)
aei = (1)(3)(2) = **6**	ceg = (2)(3)(6) = **36**
bfg = (0)(−3)(6) = **0**	afh = (1)(−3)(−1) = **3**
cdh = (2)(5)(−1) = **−10**	bdi = (0)(5)(2) = **0**

> det D = (6 + 0 − 10) − (36 + 3 + 0) = (−4) − (39) = **−43**

Double-check with cofactor expansion on row 1: 1·|3 −3; −1 2| − 0·(...) + 2·|5 3; 6 −1| = 1(6 − 3) + 2(−5 − 18) = 3 − 46 = −43 ✓

FINAL ANSWER

det(D) = **−43**

9. B · A

B is 2×**3** A is **2**×2 → Inner numbers do NOT match (3 ≠ 2) ✗ **NOT DEFINED**

ANSWER

B·A is NOT defined. B has 3 columns but A has only 2 rows. (Notice A·B in #1 worked, but B·A doesn't. Order matters!)

10. $A^T \cdot B$

Step 1: Transpose A (rows become columns):

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Step 2: Multiply $A^T \cdot B$:

A^T is 2×**2** B is **2**×3 → Inner numbers match (2 = 2) ✓ Answer size = **2×3**

Entry	What to multiply	Work	=
c_{11}	Row 1 of A^T · Col 1 of B	(1)(5) + (3)(6) = 5 + 18	**23**
c_{12}	Row 1 of A^T · Col 2 of B	(1)(3) + (3)(2) = 3 + 6	**9**
c_{13}	Row 1 of A^T · Col 3 of B	(1)(2) + (3)(1) = 2 + 3	**5**
c_{21}	Row 2 of A^T · Col 1 of B	(2)(5) + (4)(6) = 10 + 24	**34**
c_{22}	Row 2 of A^T · Col 2 of B	(2)(3) + (4)(2) = 6 + 8	**14**
c_{23}	Row 2 of A^T · Col 3 of B	(2)(2) + (4)(1) = 4 + 4	**8**

FINAL ANSWER

$$A^T \cdot B = \begin{bmatrix} 23 & 9 & 5 \\ 34 & 14 & 8 \end{bmatrix}$$

9 Determinant of a Matrix

💡 **THE IDEA (IN PLAIN WORDS)**

The **determinant** is **one single number** computed from a **square** matrix. It is written det A or |A|. Think of it as the **scaling factor**: how much the matrix stretches or shrinks area/volume.

Most important use: if det A = 0, the matrix has **NO inverse**.

A. 2×2 determinant: "ad − bc"

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Multiply the **main diagonal** (↘), then subtract the product of the **other diagonal** (↙).

$$\text{Example: } \det \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} = (4)(6) - (7)(2) = 24 - 14 = \mathbf{10}$$

B. 3×3 determinant: Sarrus / diagonal method (from the slides)

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\det A = aei + bfg + cdh - ceg - afh - bdi$$

How to remember it: Copy the first two columns to the right side of the matrix:

a	b	c	a	b
d	e	f	d	e
g	h	i	g	h

↘ **3 down-diagonals, ADD:** aei, bfg, cdh

↗ **3 up-diagonals, SUBTRACT:** gec, hfa, idb (same as ceg, afh, bdi)

Worked example: the slides' matrix A (used again in the Inverse section)

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

Down diagonals (+)	Up diagonals (−)
aei = (3)(0)(1) = **0**	ceg = (2)(0)(0) = **0**
bfg = (0)(−2)(0) = **0**	afh = (3)(−2)(1) = **−6**

cdh = (2)(2)(1) = **4**	bdi = (0)(2)(1) = **0**

det A = (0 + 0 + 4) − (0 − 6 + 0) = (4) − (−6) = **10**

FINAL ANSWER

det A = **10**

C. Cofactor expansion (for 4×4 or bigger, and it also works for 3×3)

Go across the **first row**. For each entry:

1. Cover its row and column. The numbers left form a smaller matrix (the **minor**).
2. Multiply the entry by the determinant of that smaller matrix.
3. Use the sign pattern **+ − + − …** and add everything.

$$|A| = a\cdot|M_{11}| - b\cdot|M_{12}| + c\cdot|M_{13}| - d\cdot|M_{14}|$$

where $\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \rightarrow M_{11} = \begin{bmatrix} f & g & h \\ j & k & l \\ n & o & p \end{bmatrix}$ $M_{12} = \begin{bmatrix} e & g & h \\ i & k & l \\ m & o & p \end{bmatrix}$

$M_{13} = \begin{bmatrix} e & f & h \\ i & j & l \\ m & n & p \end{bmatrix}$ $M_{14} = \begin{bmatrix} e & f & g \\ i & j & k \\ m & n & o \end{bmatrix}$

📝 **NOTE ABOUT THE SLIDES**

On slide 26, the minor for **b** is typed with *f* and *n* in the first column. The correct minor for b (cover row 1 and column 2) is **[e g h; i k l; m o p]**, as shown above.

Bonus worked 4×4 example

$$M = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

+ a = 2: cover row 1 and column 1

$$\begin{bmatrix} \mathbf{2} & 1 & 0 & 3 \\ 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 0 & 2 & 1 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix} = -8$$

Down diagonals (+)	Up diagonals (−)
aei = (0)(1)(1) = **0**	ceg = (1)(1)(2) = **2**
bfg = (2)(0)(2) = **0**	afh = (0)(0)(0) = **0**
cdh = (1)(3)(0) = **0**	bdi = (2)(3)(1) = **6**

> $\det M_{11}$ = (0 + 0 + 0) − (2 + 0 + 6) = (0) − (8) = **−8**

> term = + (2) × (−8) = **−16**

− b = 1: cover row 1 and column 2

$$\begin{bmatrix} 2 & \mathbf{1} & 0 & 3 \\ 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 0$$

Down diagonals (+)	Up diagonals (−)
aei = (1)(1)(1) = **1**	ceg = (1)(1)(1) = **1**
bfg = (2)(0)(1) = **0**	afh = (1)(0)(0) = **0**
cdh = (1)(0)(0) = **0**	bdi = (2)(0)(1) = **0**

> $\det M_{12}$ = (1 + 0 + 0) − (1 + 0 + 0) = (1) − (1) = **0**

> term = − (1) × (0) = **0**

+ c = 0: cover row 1 and column 3

$$\begin{bmatrix} 2 & 1 & \mathbf{0} & 3 \\ 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

> term = + (0) × (0) = **0**

− d = 3: cover row 1 and column 4

$$\begin{bmatrix} 2 & 1 & 0 & \mathbf{3} \\ 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 0 & 2 \\ 0 & 3 & 1 \\ 1 & 2 & 0 \end{vmatrix} = -8$$

Down diagonals (+)	Up diagonals (−)
aei = (1)(3)(0) = **0**	ceg = (2)(3)(1) = **6**
bfg = (0)(1)(1) = **0**	afh = (1)(1)(2) = **2**
cdh = (2)(0)(2) = **0**	bdi = (0)(0)(0) = **0**

> $\det M_{14}$ = (0 + 0 + 0) − (6 + 2 + 0) = (0) − (8) = **−8**

> term = − (3) × (−8) = **24**

> det M = (−16) + (0) + (0) + (24) = **8**

FINAL ANSWER

det M = **8**

TIP

Choose a row or column with **many zeros**. A zero entry makes its whole term 0, so you can skip it. (Here, c = 0, so its term vanished immediately.)

10 Inverse of a Matrix

💡 **THE IDEA (IN PLAIN WORDS)**

The **inverse** A^{-1} is the matrix that **"undoes"** A. Just like 5 × ⅕ = 1 for numbers, for matrices: $\mathbf{A \cdot A^{-1} = A^{-1} \cdot A = I}$ (the identity matrix).

Only **square** matrices with **det ≠ 0** have an inverse.

A. The 2×2 shortcut formula

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Recipe: ① **Swap** a and d. ② Change the **signs** of b and c. ③ **Divide** everything by the determinant (ad − bc).

Slide example

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

Step 1: det A = (4)(6) − (7)(2) = 24 − 14 = **10** (not zero, so an inverse exists ✓)

Step 2: Swap 4 and 6, and negate 7 and 2:

$$\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

Step 3: Divide by 10:

FINAL ANSWER

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

Check: $\mathbf{A \cdot A^{-1}}$ should equal I.

Entry	Work	=
(1,1)	(4)(0.6) + (7)(−0.2) = 2.4 − 1.4	**1**
(1,2)	(4)(−0.7) + (7)(0.4) = −2.8 + 2.8	**0**
(2,1)	(2)(0.6) + (6)(−0.2) = 1.2 − 1.2	**0**
(2,2)	(2)(−0.7) + (6)(0.4) = −1.4 + 2.4	**1**

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \checkmark$$

B. For 3×3 or bigger, there are two methods

1. **Elementary Row Operations** (Gauss-Jordan method)
2. **Minors, Cofactors, and Adjugate**

Both methods give the **same answer**. Below, both are done on the slides' matrix:

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

Method 1: Elementary Row Operations (Gauss-Jordan)

💡 **THE IDEA (IN PLAIN WORDS)**

Put A and I side by side: **[A | I]**. Use row operations until the LEFT side becomes I. Whatever the RIGHT side turns into is A^{-1}: **[A | I] → [I | A^{-1}]**.

The 3 allowed row operations:

1. **Swap** two rows (R1 ↔ R2)
2. **Multiply** a row by a non-zero number (R1 → ½R1)
3. **Add/subtract** a multiple of another row (R2 → R2 − 3/2·R1)

Always do the operation on the **whole row**, including the right side of the bar. The highlighted row is the one that changed.

Start: [A | I]

$$\left[\begin{array}{ccc|ccc} 3 & 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Step 1: **R1 ↔ R2** (Swap row 1 and row 2 (so we start with a simpler row).)

> Row 1 and row 2 trade places.

$$\left[\begin{array}{ccc|ccc} \mathbf{2} & \mathbf{0} & \mathbf{-2} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{3} & \mathbf{0} & \mathbf{2} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Step 2: **R2 → R2 − (3/2)R1** (Make the entry below the first pivot equal to 0.)

> New R2 = [3, 0, 2 | 1, 0, 0] − 3/2 × [2, 0, −2 | 0, 1, 0] = [0, 0, 5 | 1, −3/2, 0]

$$\left[\begin{array}{ccc|ccc} 2 & 0 & -2 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{5} & \mathbf{1} & \mathbf{-\frac{3}{2}} & \mathbf{0} \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Step 3: **R2 ↔ R3** (Swap rows 2 and 3 so the '1' in column 2 goes to row 2.)

> Row 2 and row 3 trade places.

$$\left[\begin{array}{ccc|ccc} 2 & 0 & -2 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{5} & \mathbf{1} & \mathbf{-\frac{3}{2}} & \mathbf{0} \end{array}\right]$$

Step 4: **R1 → (1/2)R1** (Make the first pivot equal to 1.)

> New R1 = 1/2 × [2, 0, −2 | 0, 1, 0] = [1, 0, −1 | 0, 1/2, 0]

$$\left[\begin{array}{ccc|ccc} \mathbf{1} & \mathbf{0} & \mathbf{-1} & \mathbf{0} & \mathbf{\frac{1}{2}} & \mathbf{0} \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 5 & 1 & -\frac{3}{2} & 0 \end{array}\right]$$

Step 5: **R3 → (1/5)R3** (Make the third pivot equal to 1.)

> New R3 = 1/5 × [0, 0, 5 | 1, −3/2, 0] = [0, 0, 1 | 1/5, −3/10, 0]

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{\frac{1}{5}} & \mathbf{-\frac{3}{10}} & \mathbf{0} \end{array}\right]$$

Step 6: **R1 → R1 + R3** (Clear the −1 above the third pivot.)

> New R1 = [1, 0, −1 | 0, 1/2, 0] + [0, 0, 1 | 1/5, −3/10, 0] = [1, 0, 0 | 1/5, 1/5, 0]

$$\left[\begin{array}{ccc|ccc} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{\frac{1}{5}} & \mathbf{\frac{1}{5}} & \mathbf{0} \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \end{array}\right]$$

Step 7: **R2 → R2 − R3** (Clear the 1 above the third pivot.)

> New R2 = [0, 1, 1 | 0, 0, 1] − [0, 0, 1 | 1/5, −3/10, 0] = [0, 1, 0 | −1/5, 3/10, 1]

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{-\frac{1}{5}} & \mathbf{\frac{3}{10}} & \mathbf{1} \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \end{array}\right]$$

FINAL ANSWER

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{3}{10} & 1 \\ \frac{1}{5} & -\frac{3}{10} & 0 \end{bmatrix} = \begin{bmatrix} 0.2 & 0.2 & 0 \\ -0.2 & 0.3 & 1 \\ 0.2 & -0.3 & 0 \end{bmatrix}$$

📝 **NOTE ABOUT THE SLIDES**

The slides label every in-between matrix as "I|A^{-1}". Only the **final** matrix is really [I | A^{-1}]; the ones before it are just steps along the way.

Method 2: Minors, Cofactors, and Adjugate

4 steps: ① Matrix of Minors → ② Matrix of Cofactors → ③ Adjugate (transpose) → ④ Multiply by 1/det.

Step ① Matrix of Minors

For each position: **cover its row and column**, then take the **2×2 determinant** (ad − bc) of the 4 numbers left. The **circled** entry is the position, and the faded ones are covered.

Minor	Cover row & column	What's left	ad − bc	=
M_{11}	$\begin{bmatrix} \mathbf{3} & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix}$	(0)(1) − (−2)(1) = 0 − (−2)	**2**
M_{12}	$\begin{bmatrix} 3 & \mathbf{0} & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix}$	(2)(1) − (−2)(0) = 2 − (0)	**2**
M_{13}	$\begin{bmatrix} 3 & 0 & \mathbf{2} \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix}$	(2)(1) − (0)(0) = 2 − (0)	**2**
M_{21}	$\begin{bmatrix} 3 & 0 & 2 \\ \mathbf{2} & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix}$	(0)(1) − (2)(1) = 0 − (2)	**−2**

M_{22}	$\begin{bmatrix} 3 & 0 & 2 \\ 2 & \mathbf{0} & -2 \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix}$	$(3)(1) - (2)(0) = 3 - (0)$	**3**
M_{23}	$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & \mathbf{-2} \\ 0 & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix}$	$(3)(1) - (0)(0) = 3 - (0)$	**3**
M_{31}	$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ \mathbf{0} & 1 & 1 \end{bmatrix}$	$\begin{vmatrix} 0 & 2 \\ 0 & -2 \end{vmatrix}$	$(0)(-2) - (2)(0) = 0 - (0)$	**0**
M_{32}	$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & \mathbf{1} & 1 \end{bmatrix}$	$\begin{vmatrix} 3 & 2 \\ 2 & -2 \end{vmatrix}$	$(3)(-2) - (2)(2) = -6 - (4)$	**−10**
M_{33}	$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & \mathbf{1} \end{bmatrix}$	$\begin{vmatrix} 3 & 0 \\ 2 & 0 \end{vmatrix}$	$(3)(0) - (0)(2) = 0 - (0)$	**0**

$$\text{Matrix of Minors} = \begin{bmatrix} 2 & 2 & 2 \\ -2 & 3 & 3 \\ 0 & -10 & 0 \end{bmatrix}$$

Step ② Matrix of Cofactors

Apply the **checkerboard sign pattern**. Where there's a **−**, flip the sign of the minor. Where there's a **+**, keep it.

$$\begin{bmatrix} 2 & 2 & 2 \\ -2 & 3 & 3 \\ 0 & -10 & 0 \end{bmatrix} \text{ with signs } \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow \begin{bmatrix} 2 & \mathbf{-2} & 2 \\ \mathbf{2} & 3 & \mathbf{-3} \\ 0 & \mathbf{10} & 0 \end{bmatrix}$$

Changed: $-(2) = -2$, $-(-2) = 2$, $-(3) = -3$, $-(-10) = 10$.

Step ③ Adjugate = transpose of the cofactor matrix

$$\begin{bmatrix} 2 & -2 & 2 \\ 2 & 3 & -3 \\ 0 & 10 & 0 \end{bmatrix} \rightarrow \text{Adj}(A) = \begin{bmatrix} 2 & 2 & 0 \\ -2 & 3 & 10 \\ 2 & -3 & 0 \end{bmatrix}$$

Step ④ Multiply by 1 / det A

From the Determinant section, det A = **10**. (Quick check with row 1 and the cofactors: 3(2) + 0(−2) + 2(2) = 10 ✓)

FINAL ANSWER

$$A^{-1} = \frac{1}{10}\begin{bmatrix} 2 & 2 & 0 \\ -2 & 3 & 10 \\ 2 & -3 & 0 \end{bmatrix} = \begin{bmatrix} 0.2 & 0.2 & 0 \\ -0.2 & 0.3 & 1 \\ 0.2 & -0.3 & 0 \end{bmatrix}$$

TIP

Same answer as Gauss-Jordan ✓. To be extra sure, check one entry of $A \cdot A^{-1}$: row 1 of A (3, 0, 2) · column 1 of A^{-1} (0.2, −0.2, 0.2) = 0.6 + 0 + 0.4 = 1 ✓

11 Practical Application: Solving a System of Equations

Solve for x, y, and z:

$$x + y + z = 6$$
$$2y + 5z = -4$$
$$2x + 5y - z = 27$$

💡 **THE IDEA (IN PLAIN WORDS)**

Write the system as one matrix equation **AX = B**, then solve **X = A^{-1}B**. It's just like solving 3x = 12 by doing x = 3^{-1}·12 = 4.

- **A** = matrix of **coefficients** (the numbers in front of x, y, z)
- **X** = column of the **variables** (x, y, z)
- **B** = column of the **constants** (the numbers after =)

Step 1: Write the coefficients (fill in the hidden ones!)

$$1x + 1y + 1z = 6$$
$$0x + 2y + 5z = -4 \quad \text{(no x means 0x)}$$
$$2x + 5y - 1z = 27 \quad \text{(−z means −1z)}$$

$$\mathbf{A}\ \mathbf{X} = \mathbf{B}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \\ 27 \end{bmatrix}$$

✅ **TIP**

Size check for X = A^{-1}B: (3×**3**)(**3**×1) → 3×1, which matches the size of X ✓

Step 2: Find A^{-1} (Minors → Cofactors → Adjugate → ÷ det)

2A. Matrix of Minors

Minor	Cover row & column	What's left	ad − bc	=
M_{11}	$\begin{bmatrix} \mathbf{1} & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 2 & 5 \\ 5 & -1 \end{vmatrix}$	(2)(−1) − (5)(5) = −2 − (25)	**−27**
M_{12}	$\begin{bmatrix} 1 & \mathbf{1} & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 0 & 5 \\ 2 & -1 \end{vmatrix}$	(0)(−1) − (5)(2) = 0 − (10)	**−10**

M_{13}	$\begin{bmatrix} 1 & 1 & \mathbf{1} \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 0 & 2 \\ 2 & 5 \end{vmatrix}$	$(0)(5) - (2)(2) = 0 - (4)$	$\mathbf{-4}$
M_{21}	$\begin{bmatrix} 1 & 1 & 1 \\ \mathbf{0} & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 5 & -1 \end{vmatrix}$	$(1)(-1) - (1)(5) = -1 - (5)$	$\mathbf{-6}$
M_{22}	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & \mathbf{2} & 5 \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}$	$(1)(-1) - (1)(2) = -1 - (2)$	$\mathbf{-3}$
M_{23}	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & \mathbf{5} \\ 2 & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix}$	$(1)(5) - (1)(2) = 5 - (2)$	$\mathbf{3}$
M_{31}	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ \mathbf{2} & 5 & -1 \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix}$	$(1)(5) - (1)(2) = 5 - (2)$	$\mathbf{3}$
M_{32}	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & \mathbf{5} & -1 \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 0 & 5 \end{vmatrix}$	$(1)(5) - (1)(0) = 5 - (0)$	$\mathbf{5}$
M_{33}	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & \mathbf{-1} \end{bmatrix}$	$\begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix}$	$(1)(2) - (1)(0) = 2 - (0)$	$\mathbf{2}$

$$\text{Matrix of Minors} = \begin{bmatrix} -27 & -10 & -4 \\ -6 & -3 & 3 \\ 3 & 5 & 2 \end{bmatrix}$$

> 📝 **NOTE ABOUT THE SLIDES**
>
> Slides 52–53 label this as "MoM $[A]^{-1}$". The matrix of minors is computed from **A itself**. The -1 only reminds you that the goal is the inverse.

2B. Matrix of Cofactors (apply + − + pattern)

$$\begin{bmatrix} -27 & -10 & -4 \\ -6 & -3 & 3 \\ 3 & 5 & 2 \end{bmatrix} \text{ with } \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow \begin{bmatrix} -27 & \mathbf{10} & -4 \\ \mathbf{6} & -3 & \mathbf{-3} \\ 3 & \mathbf{-5} & 2 \end{bmatrix}$$

Changed: $-(-10) = 10$, $-(-6) = 6$, $-(3) = -3$, $-(5) = -5$.

2C. Adjugate (transpose the cofactors)

$$\begin{bmatrix} -27 & 10 & -4 \\ 6 & -3 & -3 \\ 3 & -5 & 2 \end{bmatrix} \rightarrow \text{Adj}(A) = \begin{bmatrix} -27 & 6 & 3 \\ 10 & -3 & -5 \\ -4 & -3 & 2 \end{bmatrix}$$

2D. Determinant of A (Sarrus)

Down diagonals (+)	Up diagonals (−)
aei = (1)(2)(−1) = **−2**	ceg = (1)(2)(2) = **4**
bfg = (1)(5)(2) = **10**	afh = (1)(5)(5) = **25**
cdh = (1)(0)(5) = **0**	bdi = (1)(0)(−1) = **0**

> det A = (−2 + 10 + 0) − (4 + 25 + 0) = (8) − (29) = **−21**

Check with row 1 × cofactors: 1(−27) + 1(10) + 1(−4) = −21 ✓

2E. A^{-1} = (1/det) × Adjugate

$$A^{-1} = \frac{1}{-21}\begin{bmatrix} -27 & 6 & 3 \\ 10 & -3 & -5 \\ -4 & -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{9}{7} & -\frac{2}{7} & -\frac{1}{7} \\ -\frac{10}{21} & \frac{1}{7} & \frac{5}{21} \\ \frac{4}{21} & \frac{1}{7} & -\frac{2}{21} \end{bmatrix}$$

Simplify each: −27/−21 = 9/7, 6/−21 = −2/7, 3/−21 = −1/7, 10/−21 = −10/21, −3/−21 = 1/7, −5/−21 = 5/21, −4/−21 = 4/21, −3/−21 = 1/7, 2/−21 = −2/21.

Step 3: Multiply X = $A^{-1}B$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{9}{7} & -\frac{2}{7} & -\frac{1}{7} \\ -\frac{10}{21} & \frac{1}{7} & \frac{5}{21} \\ \frac{4}{21} & \frac{1}{7} & -\frac{2}{21} \end{bmatrix} \begin{bmatrix} 6 \\ -4 \\ 27 \end{bmatrix}$$

Var	Row of A^{-1} · B (use 21 as common denominator)	=
x	(9/7)(6) + (−2/7)(−4) + (−1/7)(27) = 54/7 + 8/7 − 27/7 = (162 + 24 − 81) / 21 = 105/21	**5**
y	(−10/21)(6) + (1/7)(−4) + (5/21)(27) = −20/7 − 4/7 + 45/7 = (−60 − 12 + 135) / 21 = 63/21	**3**

z	(4/21)(6) + (1/7)(−4) + (−2/21)(27) = 8/7 − 4/7 − 18/7 = (24 − 12 − 54) / 21 = −42/21	**−2**

FINAL ANSWER

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} \rightarrow \mathbf{x = 5,\ y = 3,\ z = -2}$$

Step 4: Always check by plugging back in

Equation	Substitute x = 5, y = 3, z = −2	OK?
x + y + z = 6	5 + 3 + (−2) = 6	✓
2y + 5z = −4	2(3) + 5(−2) = 6 − 10 = −4	✓
2x + 5y − z = 27	2(5) + 5(3) − (−2) = 10 + 15 + 2 = 27	✓

12 Cheat Sheet: Everything on One Page

Topic	Rule	Requirement
Size	**rows × columns** (RC). Entry a_{ij} = row i, column j	n/a
Scalar c·A	Multiply **every** entry by c	Any matrix
A ± B	Add/subtract matching positions	**Same size**
Transpose A^T	Rows ↔ columns; m×n → n×m	Any matrix
A·B	Entry (i,j) = row i of A · column j of B (multiply pairs, then add)	(m×**n**)(**n**×p) → m×p
A^2	A·A (NOT squaring each entry)	**Square** only
det 2×2	**ad − bc**	Square
det 3×3	**aei + bfg + cdh − ceg − afh − bdi**	Square
det 4×4+	Expand along row 1: $a\|M_{11}\| - b\|M_{12}\| + c\|M_{13}\| - d\|M_{14}\|$	Square
Inverse 2×2	(1/det) × [d −b; −c a]: swap a,d; negate b,c	Square, det ≠ 0
Inverse 3×3 (Gauss-Jordan)	[A | I] → row operations → [I | A^{-1}]	Square, det ≠ 0
Inverse 3×3 (Adjugate)	Minors → Cofactors (+−+) → Transpose → × 1/det	Square, det ≠ 0
System of equations	AX = B → **$X = A^{-1}B$**	A invertible

Sign pattern for cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \quad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

Top-left is always +, then alternate.

! Common Mistakes to Avoid

1. **Mixing up rows and columns.** Size is ROWS × COLUMNS. A 2×3 matrix has 2 rows.
2. **Multiplying position-by-position.** Matrix multiplication is row × column (dot product), not entry × entry.
3. **Forgetting to check sizes first.** Always write (m×n)(n×p). If the middle numbers differ, it's NOT defined.
4. **Assuming AB = BA.** Order matters. One can be defined while the other isn't (Problem Set 5 #1 vs #9).
5. **Squaring each entry for A^2.** A^2 = A·A.
6. **Sign errors with negatives.** Put negatives in parentheses: (−3)(−4) = +12, and 1 − (−9) = 10.
7. **Forgetting the + − + pattern** when turning minors into cofactors.
8. **Forgetting to transpose** (the adjugate step) before dividing by the determinant.
9. **Dividing by the wrong sign of det.** If det = −21, every entry of the adjugate gets divided by −21, so signs flip.
10. **Doing a row operation on only half the row** in Gauss-Jordan. Apply it to both sides of the bar.

? Self-Test (answers at the bottom, so cover them first!)

1. What is the size of the product of a 2×3 matrix and a 3×4 matrix?

2. Compute $3 \cdot \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}$

3. Find $\det \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

4. Find the transpose of $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

5. Compute $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

6. Find the inverse of $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$

7. Is $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ defined?

8. Find $\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$

ANSWERS

1. **2×4** (inner 3 = 3 ✓, outer numbers 2 and 4)
2. $\begin{bmatrix} 3 & -6 \\ 0 & 12 \end{bmatrix}$
3. (3)(2) − (5)(1) = 6 − 5 = **1**
4. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (a 3×1 column matrix)
5. Row 1: (1)(3) + (2)(4) = 11; Row 2: (0)(3) + (1)(4) = 4 → $\begin{bmatrix} 11 \\ 4 \end{bmatrix}$
6. det = 6 − 5 = 1 → swap & negate: $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$
7. **No.** (3×**2**)(**3**×2): 2 ≠ 3, not square
8. aei + bfg + cdh − ceg − afh − bdi = 0 + 40 + 0 − 15 − 24 − 0 = **1**

Good luck, Seb! Kaya mo 'yan! 💪